

Generalized Estimating Equations (일반화 추정 방정식)

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Today's Topics

Generalized Estimating Equations

- Preliminary Model
- Normal Linear Model
- Generalized Linear Model
- Quasi-likelihood

Generalized Estimating Equations

GEEs

They are used to model correlated data from longitudinal / repeated measures studies and clustered/ multilevel studies

GEEs are extension of generalized linear model to analyze correlated data

Preliminary Model

Marginal model

Marginal models are appropriate when inferences about the population-average are the focus

For example, in a clinical trial the average difference between control and treatment is most important

Example

What is the age-specific prevalence of respiratory infection in children?

Is the prevalence of respiratory infection greater in the sub-population of children with vitamin A deficiency?

How does the association of vitamin A deficiency and respiratory infection change with age?

Preliminary Model (Cont'd)

GEE

Liang and Zeger (1986, *Biometrika*)

Zeger, Linag and Self (1985, *Biometrika*)

Extension of GLM using quasi-likelihood estimation

Generalized linear model:

$$g(\mu_{ij}) = x_{ij}^T \beta$$

where $g(\cdot)$ is a link function

Notations

Repeated measurements: $y_{ij}, i = 1, \dots, m$ subject; $j = 1, \dots, n_i$ times for subject i

Clustered data: $y_{ij}, i = 1, \dots, m$ clusters; $j = 1, \dots, n_i$ measurements within cluster i

Use "unit" for subject or cluster

Vector of measurements for unit i :

$$Y_i = \begin{bmatrix} y_{i1} \\ y_{i2} \\ \vdots \\ y_{in_i} \end{bmatrix}, n = \sum_{i=1}^m n_i$$

Vector of measurements for all units

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_m \end{bmatrix}$$

Normal Linear Model: Estimation

For unit i :

$$\mathbf{E}(\mathbf{y}_i) = \mu_i = \mathbf{X}_i \boldsymbol{\beta}; \quad \mathbf{y}_i \sim \mathcal{N}(\mu_i, \mathbf{V}_i)$$

$\mathbf{X}_i$: $n_i \times p$ design matrix

$\boldsymbol{\beta}$: $p \times 1$ parameter vector

$\mathbf{V}_i$: $n_i \times n_i$ variance-covariance matrix,

For example, $\mathbf{V}_i = \sigma^2 \mathbf{I}$ if measurements are independent

For all units:

$$\mathbf{E}(\mathbf{y}) = \boldsymbol{\mu} = \mathbf{X} \boldsymbol{\beta}, \quad \mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{V})$$

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_m \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \\ \vdots \\ \mathbf{X}_m \end{bmatrix}, \quad \mathbf{V} = \begin{bmatrix} \mathbf{V}_1 & 0 & 0 \\ 0 & \ddots & \\ 0 & & \mathbf{V}_m \end{bmatrix}$$

This $\mathbf{V}$ is suitable if the units are independent

Normal Linear Model: Estimation (Cont'd)

We want to estimate β and V

Use

$$\begin{aligned}\text{log-likelihood function } \ell &\propto (y - \mu)^T V^{-1} (y - \mu) \\ \text{Score} = U(\beta) &= \frac{\partial \ell}{\partial \beta} = X^T V^{-1} (y - \mu) \\ &= \sum X_i^T V_i^{-1} (y_i - X_i \beta) = 0\end{aligned}$$

Solve this set of score equations to estimate β

Generalized Linear Model

GLM

Relate the marginal response to a linear combination of the covariates

$$g(\mu_{ij}) = x_{ij}^T \beta$$

Describe the variance of y_{ij} as a function of the mean

$$\text{var}(y_{ij}) \equiv V(\mu_{ij}) = \phi v(\mu_{ij})$$

Link and variance function

Distribution	$g(\mu_{ij})$	$v(\mu_{ij})$	ϕ
Normal	μ_{ij}	1	ϕ
Binary	$\log [\mu_{ij}/(1 - \mu_{ij})]$	$\mu_{ij} (1 - \mu_{ij})$	1
Poisson	$\log (\mu_{ij})$	μ_{ij}	1

Extension to GEE

y_{ij} s (elements of $\mathbf{y}_i$) are not necessarily normal (e.g., Poisson, binomial)

$$E(Y_{ij}) = \mu_{ij}$$

$g(\mu_{ij}) = \eta_{ij} = X_i\beta$; g is the link function

$$\text{Score} = \mathbf{U}(\boldsymbol{\beta}) = \sum \mathbf{D}_i^T \mathbf{V}_i^{-1}(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$$

where $\mathbf{D}_i$ is a matrix of derivatives with elements

$$\frac{\partial \mu_i}{\partial \beta_k} = \frac{\partial \mu_i}{\partial \eta_k} x_{ik}$$

and $\mathbf{V}_i$ is diagonal with elements $\text{var}(Y_{ij})$ (if link is identity, then $\mathbf{D}_i = \mathbf{X}_i$)

Quasi-Likelihood

For exponential family distribution

Exponential family

$$l = \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)$$

$$U = \frac{\partial l}{\partial \mu}$$

$$\frac{\partial l}{\partial \mu} = \frac{y - b'(\theta)}{a(\phi)} \frac{\partial \theta}{\partial \mu} = \frac{y - \mu}{a(\phi)b''(\theta)} = \frac{y - \mu}{a(\phi)v(\mu)}$$

Property of U

$$E(U) = 0$$

$$E(U^2) = -E\left(\frac{\partial^2 l}{\partial \mu^2}\right) = \frac{1}{a(\phi)v(\mu)}$$

Quasi-Likelihood (Cont'd)

Quasi-likelihood (QL)

$$Q(\mu; Y) = \int_Y^{\mu} \frac{Y-t}{\phi v(t)} dt$$

Property of Q

For fixed ϕ , it behaves like a log likelihood function

For many common distributions, a QL is identical to a log likelihood function

However, it is likely that a QL exists but a log likelihood does not, since one may specify an arbitrary variance function

$$\frac{\partial Q}{\partial \mu} = \frac{Y - \mu}{a(\phi)v(\mu)}$$

Quasi-Likelihood (Cont'd)

Quasi-deviance

$$D(\mu; Y) = -2\phi Q(\mu; Y) = 2 \int_{\mu}^Y \frac{Y-t}{\phi v(t)} dt$$

Then, $D(\mu; Y) \geq 0$ and $D(\mu; Y)$ does not depend on ϕ

Quasi score function for β

$$U(\beta) = \frac{1}{\phi} \mathbf{D}^t \mathbf{V}^{-1} (\mathbf{y} - \boldsymbol{\mu}), \mathbf{D} = \frac{\partial \boldsymbol{\mu}}{\partial \beta} \text{ \& } \mathbf{V} = \mathbf{v}(\boldsymbol{\mu})$$

Quasi information matrix for β

$$I(\beta) = -E \left\{ \frac{\partial U(\beta)}{\partial \beta} \right\} = \frac{1}{\phi} \mathbf{D}^t \mathbf{V}^{-1} \mathbf{D}$$

Quasi-Likelihood (Cont'd)

Theorem (Wedderburn 1974)

Under some regularity conditions, the maximum quasi-likelihood estimator $\hat{\beta}$ which satisfies $U(\hat{\beta}) = 0$, is consistent and asymptotically normal

$$\hat{\beta} \xrightarrow{P} \beta$$

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{D} N(0, \mathbf{I}^{-1}(\beta))$$

If variance is mis-specified, what happens?

Quasi-Likelihood (Cont'd)

Theorem

Within the class of linear estimating function $\mathbf{H}^t(\mathbf{y} - \boldsymbol{\mu}(\theta))$

$$U(\beta) = \mathbf{D}^t \mathbf{V}^{-1} (\mathbf{y} - \boldsymbol{\mu}(\theta))$$

is optimal

Optimality means

$$\text{var}(\hat{\theta}) - \text{var}(\hat{\theta}_0) \geq 0 \text{ where } \hat{\theta} \text{ and } \hat{\theta}_0 \text{ are solutions of } \mathbf{H}^t(\mathbf{y} - \boldsymbol{\mu}(\hat{\theta})) = 0 \\ \& \mathbf{D}^t \mathbf{V}^{-1} (\mathbf{y} - \boldsymbol{\mu}(\hat{\theta}_0)) = 0$$

Optimal Linear Estimating Equation

Proof It can be proven by the Taylor expansion:

Let $h(\beta) = \mathbf{H}^t(\mathbf{y} - \boldsymbol{\mu}(\beta))$. Then by Taylor expansion, we have

$$0 = h(\hat{\beta}) \approx h(\beta) - \mathbf{H}^t \mathbf{D}(\hat{\beta} - \beta)$$

Therefore we have

$$\sqrt{m}(\hat{\beta} - \beta) = (m^{-1} \mathbf{H}^t \mathbf{D})^{-1} m^{-1/2} \mathbf{H}^t (\mathbf{y} - \boldsymbol{\mu}(\hat{\beta})) + o_P(1) \xrightarrow{D} N(0, \Sigma)$$

$$\text{where } \Sigma = \lim_{m \rightarrow \infty} \{ (n^{-1} \mathbf{H}^t \mathbf{D})^{-1} (m^{-1} \mathbf{H}^t \mathbf{V} \mathbf{H}) [(n^{-1} \mathbf{H}^t \mathbf{D})^{-1}]^t \}$$

$$\sqrt{m}(\hat{\beta}_0 - \beta) \xrightarrow{D} N(0, \Sigma_0), \Sigma_0 = \lim_{m \rightarrow \infty} \{ (m^{-1} \mathbf{D}^t \mathbf{V}^{-1} \mathbf{D})^{-1} \}$$

Thus

$$\Sigma_0^{-1} - \Sigma^{-1} = \lim_{m \rightarrow \infty} \{ n^{-1} \mathbf{D}^t \{ \mathbf{V}^{-1} - \mathbf{H}(\mathbf{H}^t \mathbf{V} \mathbf{H})^{-1} \mathbf{H}^t \} \mathbf{D} \} \geq 0$$

✂ It should be reminded that

$$\mathbf{V}^{-1} - \mathbf{A}(\mathbf{A}^t \mathbf{V}^{-1} \mathbf{A})^{-1} \mathbf{A}^t = \mathbf{V}^{-1} \mathbf{X}(\mathbf{X}^t \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^t \mathbf{V}^{-1}$$